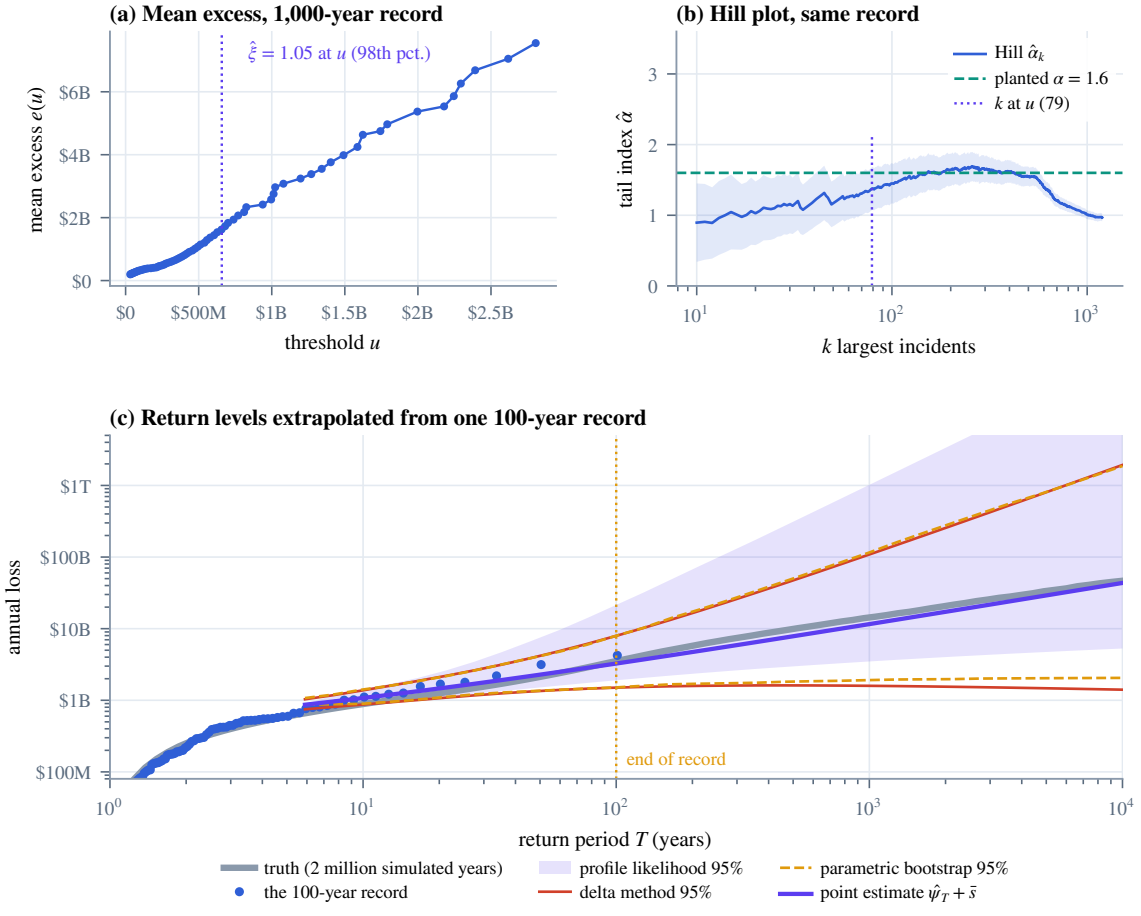

(a) Mean excess, 1,000-year record
$\hat{\xi}$ = 1.05 at u (98th pct.)
mean excess $e(u)$
$6B
$4B
$2B
$0
$0
$500M
$1B
$1.5B
$2B
$2.5B
threshold u
(b) Hill plot, same record
Hill $\hat{\alpha}_k$
planted α = 1.6
k at u (79)
tail index $\hat{\alpha}$
3
2
1
0
10^1
10^2
10^3
k largest incidents
(c) Return levels extrapolated from one 100-year record
annual loss
$1T
$100B
$10B
$1B
$100M
end of record
10^0
10^1
10^2
10^3
10^4
return period T (years)
truth (2 million simulated years)
the 100-year record
profile likelihood 95%
delta method 95%
parametric bootstrap 95%
point estimate $\hat{\psi}_T + \bar{s}$